

30 Sample Practical Programs

Category A: Basic Syntax & Arithmetic (Programs 1-10)

1. Hello World

```
<?php
echo "Hello World!";
?>
```

2. Simple Addition of Two Numbers

```
<?php
$a = 15;
$b = 25;
$sum = $a + $b;
echo "The sum is: " . $sum;
?>
```

3. Swap Two Numbers Using a Third Variable

```
<?php
$a = 5;
$b = 10;
$temp = $a;
$a = $b;
$b = $temp;
echo "After swapping: a = $a, b = $b";
?>
```

4. Swap Two Numbers Without a Third Variable

```
<?php
$a = 10;
$b = 5;
$a = $a + $b; // 15
$b = $a - $b; // 10
$a = $a - $b; // 5
echo "After swapping: a = $a, b = $b";
?>
```

5. Calculate Area of a Rectangle

```
<?php
$length = 20;
$width = 10;
$area = $length * $width;
echo "Area of Rectangle: " . $area;
?>
```

6. Calculate Area of a Circle

```
<?php
define("PI", 3.14159);
$radius = 7;
$area = PI * $radius * $radius;
echo "Area of Circle: " . $area;
?>
```

7. Convert Celsius to Fahrenheit

```
<?php
$celsius = 32;
$fahrenheit = ($celsius * 9/5) + 32;
echo "$celsius°C is equal to $fahrenheit°F";
?>
```

8. Find Length of a String

```
<?php
$str = "Computer Science";
echo "The length of the string is: " . strlen($str);
?>
```

9. Count Number of Words inside a String

```
<?php
$str = "Welcome to PHP web programming basics.";
echo "Word count: " . str_word_count($str);
?>
```

10. Reverse a String

```
<?php
$str = "Surat";
echo "Reversed string: " . strrev($str);
?>
```

Category B: Conditional Logic & If-Else (Programs 11-20)

11. Check if a Number is Even or Odd

```
<?php
$num = 44;
if($num % 2 == 0) {
    echo "$num is Even";
} else {
    echo "$num is Odd";
}
?>
```

12. Check if a Person is Eligible to Vote

```
<?php
$age = 19;
if($age >= 18) {
    echo "Eligible to vote.";
} else {
    echo "Not eligible to vote.";
}
?>
```

13. Find the Largest of Two Numbers

```
<?php
$x = 45;
$y = 90;
if($x > $y) {
    echo "$x is greater than $y";
} else {
    echo "$y is greater than $x";
}
?>
```

14. Find the Largest of Three Numbers

```
<?php
$a = 10; $b = 25; $c = 15;
if($a >= $b && $a >= $c) {
    echo "$a is the largest number.";
} elseif($b >= $a && $b >= $c) {
    echo "$b is the largest number.";
} else {
    echo "$c is the largest number.";
}
?>
```

15. Check if a Year is a Leap Year

```
<?php
$year = 2024;
if(($year % 400 == 0) || ($year % 4 == 0 && $year % 100 != 0)) {
    echo "$year is a Leap Year.";
} else {
    echo "$year is not a Leap Year.";
}
?>
```

16. Check if a Number is Positive, Negative, or Zero

```
<?php
$num = -7;
if($num > 0) {
    echo "$num is Positive";
} elseif($num < 0) {
    echo "$num is Negative";
} else {
    echo "The number is Zero";
}
?>
```

17. Simple Calculator Using Switch Case

```
<?php
$num1 = 12;
$num2 = 4;
$operator = '/';
```

```

switch($operator) {
    case '+': echo $num1 + $num2; break;
    case '-': echo $num1 - $num2; break;
    case '*': echo $num1 * $num2; break;
    case '/': echo $num1 / $num2; break;
    default: echo "Invalid Operator";
}
?>

```

18. Check if a Character is a Vowel or Consonant

```

<?php
$char = 'e';
switch(strtolower($char)) {
    case 'a': case 'e': case 'i': case 'o': case 'u':
        echo "$char is a Vowel.";
        break;
    default:
        echo "$char is a Consonant.";
}
?>

```

19. Student Grading Scheme

```

<?php
$score = 85;
if($score >= 90) echo "Grade A";
elseif($score >= 75) echo "Grade B";
elseif($score >= 50) echo "Grade C";
else echo "Fail";
?>

```

20. Check if a Number Lies Within a Specific Range (10 to 50)

```

<?php
$num = 27;
if($num >= 10 && $num <= 50) {
    echo "$num is within the range [10, 50]";
} else {
    echo "$num is out of range";
}
?>

```

Category C: Iteration & Loops (Programs 21-30)

21. Print 1 to 10 Using a For Loop

```
<?php
for($i = 1; $i <= 10; $i++) {
    echo $i . " ";
}
?>
```

22. Print 10 to 1 in Reverse Using a While Loop

```
<?php
$i = 10;
while($i >= 1) {
    echo $i . " ";
    $i--;
}
?>
```

23. Print the Multiplication Table of 5

```
<?php
$stable = 5;
for($i = 1; $i <= 10; $i++) {
    $res = $stable * $i;
    echo "$stable x $i = $res <br>";
}
?>
```

24. Calculate Factorial of a Number

```
<?php
$num = 5;
$factorial = 1;
for($i = 1; $i <= $num; $i++) {
    $factorial *= $i;
}
echo "Factorial of $num is: " . $factorial;
?>
```

25. Fibonacci Series up to 10 Terms

```
<?php
$n = 10;
$n1 = 0; $n2 = 1;
echo "Fibonacci Series: " . $n1 . " " . $n2 . " ";
for($i = 3; $i <= $n; $i++) {
    $n3 = $n1 + $n2;
    echo $n3 . " ";
    $n1 = $n2;
    $n2 = $n3;
}
?>
```

26. Sum of First 100 Natural Numbers

```
<?php
$sum = 0;
for($i = 1; $i <= 100; $i++) {
    $sum += $i;
}
echo "The sum is: " . $sum;
?>
```

27. Display All Even Numbers Between 1 and 20

```
<?php
for($i = 1; $i <= 20; $i++) {
    if($i % 2 == 0) {
        echo $i . " ";
    }
}
?>
```

28. Count Digits of an Integer

```
<?php
$num = 14526;
$count = 0;
while($num != 0) {
    $num = (int)($num / 10);
}
```

```
    $count++;  
}  
echo "Total digits: " . $count;  
?>
```

29. Sum of Digits of a Number

```
<?php  
$num = 1234;  
$sum = 0;  
while($num > 0) {  
    $rem = $num % 10;  
    $sum += $rem;  
    $num = (int)($num / 10);  
}  
echo "Sum of digits: " . $sum;  
?>
```

30. Simple Nested Loop Star Pattern

```
<?php  
for($i = 1; $i <= 5; $i++) {  
    for($j = 1; $j <= $i; $j++) {  
        echo "* ";  
    }  
    echo "<br>";  
}  
?>
```